# A Semi-Soft Label-Guided Network With Self-Distillation for SAR Inshore Ship Detection

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***Abstract*— With the soaring development of deep learning (DL) mechanisms in recent years, convolution neural network (CNN)-based methods have been extensively investigated to achieve high accuracy of ship detection in synthetic aperture radar (SAR) images. However, existing CNN-based SAR ship detection methods still suffer from challenges in complex inshore scenarios due to the strong interference therein. To tackle this issue, a novel semi-soft label-guided network based on self-distillation (SD) for SAR ship detection ($S^2$LSDNet) is proposed in this article. First, different from the existing CNN-based detectors to extract features from the image domain only under the guidance of a one-hot label, an efficient SD training strategy is devised to extract semi-soft label information to boost the inshore ship detection accuracy. Second, an angle-related and balanced intersection-over-union (ArBIoU) loss is developed to enhance the inshore ship positioning performance by using the adaptive weights of center point bias and the aspect ratio difference. Experiments on the open SAR ship detection datasets demonstrate the effectiveness and superiority of the proposed method compared with the existing state-of-the-art approaches, especially in inshore scenes.**

***Index Terms*— Complex background, deep learning (DL), ship detection, synthetic aperture radar (SAR).**

## I. Introduction

THE task of ship detection has attracted much attention in marine surveillance applications, including resource exploitation, fishery service, and traffic management [2], [3]. Owing to the all-day and all-weather working capability, synthetic aperture radar (SAR) has been widely used for the task of ship target detection [4].

Driven by the advantages of automatic feature extraction and excellent image representation, the convolution neural network (CNN)-based approaches have made remarkable progress in the field of SAR ship detection and other remote sensing applications [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. For example, major studies [11], [12], [13] devise the scattering point acquisition modules as the crucial components for ship detection according to the electromagnetic scattering property of SAR images. Since a few scattering characteristics are not exclusive to the ship category, the detection results may be not as expected in dense and complex inshore scenarios with clutter interferences. To highlight ship targets in complex inshore scenarios, the attention mechanism has been a paradigm to suppress inshore interferences and reach decent accuracy via increasing the weights of vital areas (including ship targets) while overlooking the irrelevant stuff [14].

Canonical CNNs deal with the target detection task using feature information from the image domain under the guidance of one-hot labels (i.e., hard labels), which are simple binary annotations for classification [15]. In contrast, the soft labels are probabilistic annotations for classification to accumulate the information on entire samples instead of a rough zero setting for negative sample label values. Therefore, hard labels are easy to obtain, but ignore the abundant information of negative samples labeled with 0. Unlike hard labels, soft labels contain abundant additional knowledge to reveal the similarities between samples and real targets in detection, the advantages of which have been theoretically demonstrated in [16]. Specifically, the features of the negative samples can be well exploited via soft labels, i.e., samples that are similar to ships but not ships in practice often show relatively high probability values for network learning, while they are all marked as 0 using hard labels similar to those background areas that are significantly different from ship targets. Therefore, the participation of soft labels can markedly improve the detection accuracy against the complex background by using "dark knowledge," which is the extra probability information of negative samples to provide the network with more learnable knowledge to obtain satisfactory detection accuracy [17]. However, traditional label-guided methods still encounter obstacles and bottlenecks in practice [18]. As shown in Fig. 1(b) and (c), hard labels lack negative sample information without blue columns, and soft labels may have information loss on the positive samples and several false annotations for negative samples inducing negative feedback to the detection. In addition, the construction of a loss function to measure the differences between predicted results and ground truths (GTs) plays an important role to improve the inshore ship detection performance [19]. Since the salient aspect ratio difference and

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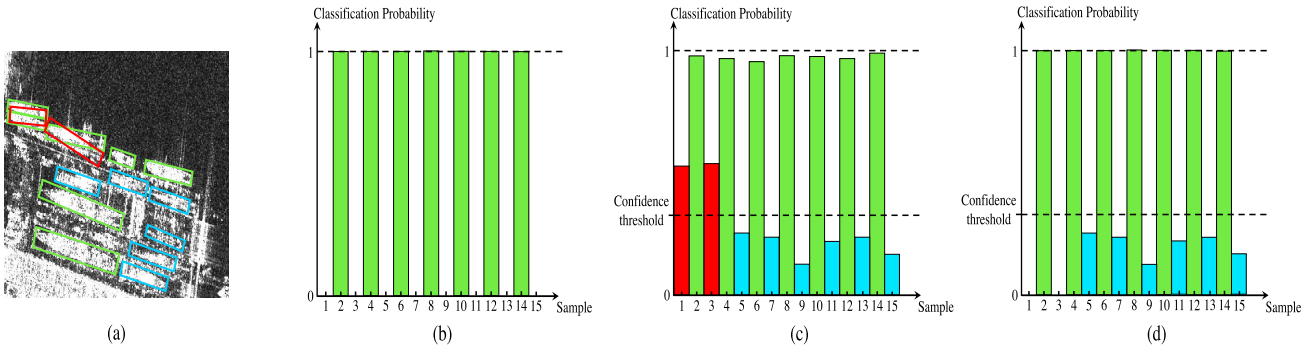



Fig. 1. Example: classification probability distributions of different labels in SAR image. The classification probability is calculated by the Softmax function, and the sample represents the bounding box in the regression process. The green, blue, and red bounding boxes in initial SAR image indicate the positive samples, the negative samples, and the mismatched annotations, respectively. Meanwhile, the green, blue, and red columns indicate the classification probability values of the corresponding boxes, respectively. (a) Initial SAR image. (b) Distribution of hard labels. (c) Distribution of soft labels. (d) Distribution of our semi-soft labels.

center point drift of inshore targets are unconsidered in the loss terms, the commonly used loss criteria are inadequate for SAR ship detection [20], such as intersection-over-union (IoU) [21], generalized IoU (GIoU) [22], and distance IoU (DIoU) [23]. To avoid the drawback, the aspect ratios and center distances between the prediction bounding boxes (BBoxes) and the corresponding GTs are introduced as extra penalty terms in loss calculation. However, existing loss metrics do not pay attention to the weight settings regarding aspect ratio and center point loss terms, leading to the dissatisfactory accuracy of ship detection in SAR images.

To address the above-mentioned issues, a semi-soft label-guided network based on self-distillation (SD) for SAR ship detection ($S^2$LSDNet) is proposed in this article, where the baseline is the dual-separation attention network (DSA-Net) [24] with a novel angle-related and balanced IoU (ArBIoU) loss metric. Instead of depending on the hard label or soft label information, our $S^2$LSDNet utilizes a new semi-soft label to strengthen the instructive knowledge of samples in complex inshore cases and enhances the ship target features via using additional ArBIoU calculation. First, we develop an ArBIoU loss to refine the feature representation of SAR ship targets by quantifying the weights of aspect ratio difference and center point offset in ship regression. Second, our method employs the DSA-Net with ArBIoU loss as a baseline. Finally, we propose a new SD strategy in which the baseline is acted as the teacher network and student network sequentially, to adaptively generate semi-soft labels that combine the advantages of soft labels and hard labels for leveraging the complete information in positive samples and "dark knowledge" from negative samples without mismatched annotations, as shown in Fig. 1(d).

The main contributions of our work can be summarized from the following aspects.

1) A semi-soft label-guided structure based on SD is proposed to enhance the ship detection performance in complex inshore scenarios. The semi-soft labels provide more learnable information and fewer false annotations compared with the commonly used hard and soft labels.
2) An ArBIoU loss is devised to achieve better regression equilibrium of the aspect ratio difference and the center point offset between predictions and GTs in ship detection.

Extensive experiments verify the superiority of our algorithm over the existing state-of-the-art (SOTA) approaches, especially in the inshore scenarios.

The remainder of this article is organized as follows. Section II briefly reviews recent research works in the field of SAR ship detection. Section III introduces the proposed method in detail. Experimental verifications of the proposed algorithm have been carried out and analyzed in Section IV. Finally, all the conclusions are presented in Section V.

## II. Related Work

In this section, brief descriptions of previous CNN-based ship detectors especially in inshore cases, and detection methods with auxiliary soft labels are provided.

### A. CNN-Based Ship Detectors in Inshore Scenarios

Compared with traditional ship detection methods based on the constant false alarm rate (CFAR), CNN-based detectors can achieve feasible performance due to their powerful feature extraction and representation capabilities [25]. Regarding the generation of prediction bounding boxes (BBoxes), existing ship detectors can be generally divided into two categories. The first category is anchor-based methods, where the predefined one-to-one proposals (two-stage model) or numerous anchors (one-stage model) were utilized to generate BBoxes of targets in regression process [26], [27], such as the faster region convolutional neural network (faster RCNN) [28] and you only look once (YOLO) v2–v4 framework [29]. The second category is anchor-free methods. Instead of generating preset anchors, anchor-free detection methods utilize the key-point estimation strategy to outline BBoxes of targets [30], [31], [32], such as CenterNet [33] and fully convolutional one-stage (FCOS) detectors [34].

Due to the cluttered background in inshore scenarios, the problem of inshore SAR ship detection received much attention in recent years. In [35], the attention mechanism was proposed to suppress inshore interferences by highlighting salient features of ship areas. Fu et al. [36] introduced

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the densely connecting convolutional block attention module (CBAM) into the pyramid network to achieve an adaptive selection of features in SAR images. Cui et al. [37] adopted the channel attention modules to suppress the background noise in complex inshore scenarios. Ma et al. [38] developed a balanced attention network based on the anchor-free framework to enhance the feature extraction capability of the network in complex backgrounds. Hu et al. [39] constructed an anchor-free structure with the spatial shuffle-group enhance (SSE) attention module to reduce false alarms caused by inshore and inland interferences. However, the existing attention mechanisms allow the network to reduce the interference of clutter samples in feature extraction only with the guide of hard labels and do not exploit substantial learnable "dark knowledge" contained by negative samples in soft labels.

Moreover, the construction of loss is critical to improving the accuracy of the detector, especially in inshore detection. Due to significant aspect ratio differences and center point offsets in ship regression, it is a common measure to introduce the aspect ratios and center distances between predictions and GTs as additional penalty terms into the regression. In [40], a skew IoU (SkewIoU) delineated the angle variation based on the geometric principle of triangulation to alleviate the negative effects of large aspect ratios in rotation detection. In [23], a complete IoU (CIoU) loss metric is devised to quantify aspect ratio differences and center point drift between the predicted BBoxes and the corresponding GTs. As a combination of SkewIoU and CIoU, SkewCIoU [24] further leverages the angular loss term into CIoU calculation to achieve accurate rotational regression for ship targets. However, none of these methods take the interaction weights of aspect ratios and center deviations into account.

### B. Soft Label-Auxiliary Ship Detector

Compared with the traditional one-hot labels, the soft labels significantly gain detection performance due to the "dark knowledge" therein. In [17], the knowledge distillation (KD) strategy was designed for the adaptive generation of soft labels to produce a thinner student network from a heavier teacher network and achieve the tradeoff between accuracy and speed. Chen et al. [41] integrated KD into the network pruning to improve the model structure without obvious performance decline in SAR ship detection. Ma et al. [42] utilized model compression and KD to accelerate the YOLOv4 network with acceptable accuracy of ship detection in SAR images. To improve the detection accuracy, as a variant of KD, the SD strategy was proposed in [43], where the same network is used as the teacher and student models. In the distillation process, the temperature is a label smoothing factor to calculate the output probability, which directly affects the distillation results and is used to exploit the negative sample information. Specifically, when the temperature value increases, the negative sample information is amplified, and the network pays more attention to the negative sample in the training process [17]. Generally, the traditional SD methods distill the knowledge in different stages or states at the same temperature (i.e., the later layers distill the previous layers [44], or the latter epochs distill the previous epochs [45]), while they make insufficient utilization of different temperature information. Despite the progress which has been made in most soft label-auxiliary methods, the drawbacks of the soft label in underutilization for positive samples and mismatched annotations for negative samples are still poorly addressed.

Motivated by the above-mentioned observations, we propose a semi-soft label-guided SAR ship detector via the SD technique in different temperatures to take advantage of both hard and soft labels. Besides, we develop a customized ArBIoU loss in BBox regression to achieve improved accuracy for inshore ship detection.

## III. Methodology

In this section, the proposed S$^2$LSDNet framework is elaborated. Specifically, the overall pipeline and architecture of the proposed network are presented in Fig. 2. Next, the DSA-Net is briefly introduced with the proposed ArBIoU loss. Finally, the SD and total loss functions are constructed in detail.

### A. Overall Scheme of Our Network

In our S$^2$LSDNet, we use the DSA-Net [24] with the ArBIoU loss as both the teacher and the student models in sequential in the SD process. The overall pipeline of the proposed method is as follows.

*Step 1:* We utilize the training data, including the images and GT labels, to pretrain an initial DSA-Net with the ArBIoU loss.

*Step 2:* The pretrained baseline, as the teacher model, infers the soft label with probability score in the high-temperature condition. In our work, the applicable high temperature is set to 20, and the specific parameter settings are discussed in later Section IV. Similar to KD, the class probability of the output in the teacher model after the Softmax layer can be calculated as

$$y_{\mathrm{ti}} = \mathrm{Softmax}\left(\frac{z_{\mathrm{tj}}}{T}\right) \tag{1}$$

where $z_{\mathrm{ti}}$ is the $i$th output value after the final fully connected layer in the teacher model, and $T$ indicates the temperature in distillation. Instead of the direct utilization of soft labels in traditional methods, we adopt semi-soft labels to both leverage the advantages of hard labels and soft labels. The proposed semi-soft label for $i$th output in the teacher model can be formulated as

$$\widehat{p_{\mathrm{li}}} = \begin{cases} 1, & \text{if label} = 1 \\ y_{\mathrm{ti}}, & \text{if label} = 0 \ \ y_{\mathrm{ti}} < T_{\mathrm{Con}} \\ 0, & \text{if label} = 0 \ \ y_{\mathrm{ti}} > T_{\mathrm{Con}} \end{cases} \tag{2}$$

where $T_{\mathrm{Con}}$ is the confidence threshold to distinguish between positive and negative samples. In addition to leveraging the "dark knowledge" in soft labels, we set the values of positive samples to 1 like the hard labels to make full use of the information in the positive sample, while modifying the erroneous annotations in the soft labels to 0 to alleviate interference from the negative sample, as presented in (2).

*Step 3:* We retrain a new DAS-Net as a student model for inference. Specifically, we use the training data (including hard

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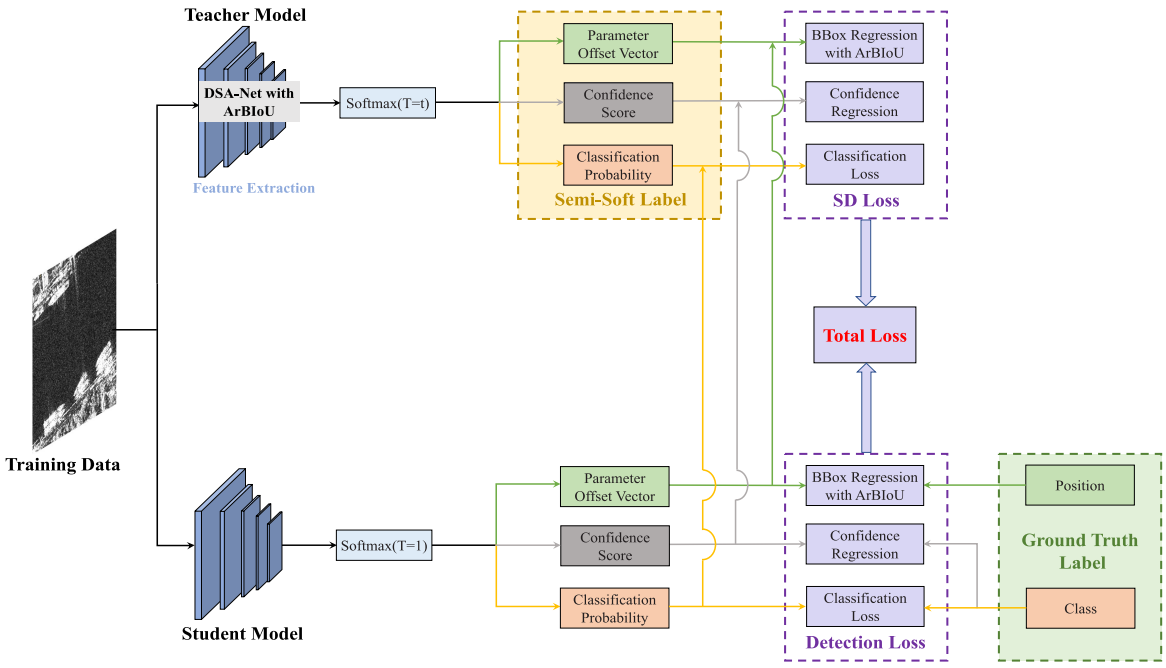



Fig. 2. Overall framework of the proposed S$^2$LSDNet. In our algorithm, we use the improved DSA-Net [24] with ArBIoU as both teacher and student models to generate semi-soft labels adaptively in different temperatures. Intuitively, the final total loss is calculated based on the SD and detection loss terms.

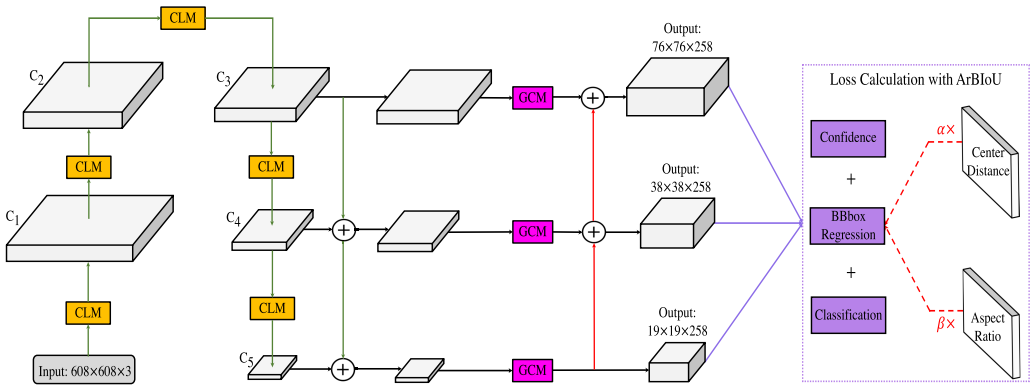



Fig. 3. Brief structure of the DSA-Net with ArBIoU loss. $C_i$ represents the $i$th feature map. The green, red, black, and purple arrows indicate downsampling, upsampling, concatenation, and loss calculation, respectively. The CLM and GCM are introduced in detail in [24]. The weight coefficients $\alpha$ and $\beta$ are explained in Section III.

labels) to calculate the detection loss while utilizing the semi-soft labels generated by the teacher model in step 2 and the soft labels generated when $T = 1$ to calculate the SD loss, and finally, obtain the optimal weights of student model based on the total loss. Note that we leverage temperature difference to ensure sufficient distillation and the retrained model shares the weights to the initial network in step 1.

*Step 4:* We set the temperature of the student model to 1 for inference.

Intuitively, different from the traditional SD techniques, we sequentially adopt the DSA-Net as the teacher and student models in different temperature conditions to achieve notable improvement for sufficient distillation. Additionally, a novel semi-soft label replaces the commonly used hard label or soft label to make full use of the information on positive and negative samples in ship detection.

### B. DSA-Net With the ArBIoU Loss

As a variant of the YOLO-series detector, the DSA-Net [24] has dual-separation attention mechanisms, i.e., the contextual location module (CLM) and the global channel module (GCM). However, the initial DSA-Net is not suitable for SAR ship detection due to the poor consideration of the characteristics in significant aspect ratio difference and center point bias in SAR ship regression. Enlightened by [9], we put forward an ArBIoU regression criterion to modify the loss calculation of DSA-Net and adapt to the SAR ship feature representation in regression by quantifying the weight association of the aspect ratio and center distance, as presented in Fig. 3. Specifically, we calculate the interrelated weights between aspect ratio offset and center point bias to balance their effects in regression and introduce an angle-related factor to outline the rotating ship targets. The ArBIoU between the

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estimated BBox and its GT is formulated as

$$\begin{aligned}&\mathrm{ArBIoU}\left(R, R^{\mathrm{GT}}\right)\\&= |\cos(\theta-\theta^{\mathrm{GT}})| \cdot \left[\mathrm{IoU}\left(R, R^{\mathrm{GT}}\right) - \alpha \cdot \frac{\rho^2\left(o, o^{\mathrm{GT}}\right)}{c^2} - \beta v\right]\end{aligned} \tag{3}$$

where $\theta$ is the angle of BBox to the horizontal axis. $o$ denotes the center point of $R$, $c$ is the diagonal length of the smallest enclosed area encompassing $R$ and $R^{\mathrm{GT}}$, $\rho(\cdot)$ denotes the Euclidean distance calculation, and the IoU calculation is defined in [20]. $v$ measures the consistency of the aspect ratio between BBox and its GT

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w}{h} - \arctan\frac{w^{\mathrm{GT}}}{h^{\mathrm{GT}}}\right) \tag{4}$$

where $w$ and $h$ are the weight and height of the BBox, respectively. The weight coefficients $\alpha$ and $\beta$ are utilized to evaluate the interaction among IoU, aspect ratios, and center distance terms, and they can be calculated by

$$\alpha = \frac{1}{\exp\left(\left|\frac{w}{h} - 1\right|\right)} \tag{5}$$

$$\beta = \frac{v}{(1-\mathrm{IoU}) + v}. \tag{6}$$

Here, $\alpha$ quantifies the interaction between center point deviation and aspect ratio difference, i.e., the aspect ratio of the ship target is approximate, the center point deviation has a greater impact on the regression, and while the difference of aspect ratio becomes larger, the impact of center point errors on the regression becomes smaller. $\beta$ establishes the association between IoU and aspect ratio, which ensures the priority of the overlap area, i.e., the larger the IoU, the greater the influence of the aspect ratio.

### C. Loss Function Construction With the SD Strategy

During the training period, the final total loss of the student model mainly comes from two portions, i.e., the detection loss based on GT labels and the SD loss based on semi-soft labels. The definition of the final loss is given by

$$L_{\mathrm{Final}} = L_{\mathrm{Dec}} + L_{\mathrm{SD}}. \tag{7}$$

Generally, the detection task consists of two parts: regression and classification. The regression subbranch is responsible for predicting the location and confidence of the bounding box in the YOLO-based framework, while the classification subbranch is responsible for predicting the category probability. In our work, the classification loss of ship detection is a binary classification issue, that is, label $= 1$ indicates the existence of a ship target, while label $= 0$ indicates the inexistence of ship targets. Hence, the binary cross entropy loss is performed as classification loss in ship detection to determine whether there is a ship. In summary, the detection loss of DSA-Net with ArBIoU can be decomposed into three parts: the BBox regression with ArBIoU, confidence regression, and classification loss. The multitask detection loss function is formulated as

$$L_{\mathrm{Dec}} = f_{\mathrm{bb}}(\boldsymbol{b}^{\boldsymbol{GT}}, \widehat{\boldsymbol{b}}) + f_{\mathrm{cs}}\left(C^{\mathrm{GT}}, \widehat{C}\right) + f_{\mathrm{cla}}\left(p^{\mathrm{GT}}, \widehat{p}\right) \tag{8}$$

$$f_{\mathrm{bb}}(\boldsymbol{b}^{\boldsymbol{GT}}, \widehat{\boldsymbol{b}}) = \frac{1}{N}\sum_{i=1}^{N} \mathrm{obj}_i \cdot |\log(\mathrm{ArBIoU})| \tag{9}$$

$$\begin{aligned}f_{\mathrm{cs}}(C^{\mathrm{GT}}, \widehat{C}) = -\Bigg[&\lambda_1 \sum_{i=1}^{N} \mathrm{obj}_i \cdot \left[C_i^{\mathrm{GT}} \log(\widehat{C}_i) + \left(1 - C_i^{\mathrm{GT}}\right) \times \log(1-\widehat{C}_i)\right]\\&+ \lambda_2 \sum_{i=1}^{N}\left(1-\mathrm{obj}_i\right) \cdot \left[C_i^{\mathrm{GT}} \log(\widehat{C}_i) + \left(1-C_i^{\mathrm{GT}}\right)\log(1-\widehat{C}_i)\right]\Bigg]\end{aligned} \tag{10}$$

$$\begin{aligned}f_{\mathrm{cla}}\left(p^{\mathrm{GT}}, \widehat{p}\right) = -\frac{1}{N}\sum_{i=1}^{\frac{N}{3}} \mathrm{obj}_i \cdot \left[p_i^{\mathrm{GT}} \log\left(\widehat{p}_i\right) + (1 - p_i^{\mathrm{GT}}) \times \log\left(1-\widehat{p}_i\right)\right]\end{aligned} \tag{11}$$

where $\widehat{\boldsymbol{b}}$ and $\widehat{p}$ are the offset vectors, including prediction parameters $(x, y, w, h, \theta)$ (i.e., the center point coordinates, width, height, and angle of BBox), and class probability calculated by Softmax layer. $\boldsymbol{b}^{\boldsymbol{GT}}$ and $p^{\mathrm{GT}}$ are the values of the corresponding parameter offset vector and probability derived from the GT. $\widehat{C}$ and $C^{\mathrm{GT}}$ are the values of predicted confidence and true confidence, respectively. The tradeoff coefficient $(\lambda_1, \lambda_2)$ is selected as $(5, 0.5)$ to ensure the priority of positive samples [24]. $N$ denotes the number of generated anchors. $\mathrm{obj}_i$ is a binary value, where $\mathrm{obj}_i = 1$ indicates the foreground in $i$th anchor, and $\mathrm{obj}_i = 0$ indicates the background areas. Similarly, subscript $i$ in the parameter represents its corresponding $i$th value. In the YOLO-based detector, each ship target matches three candidate anchors, so the $N/3$ in classification loss represents that the calculation is only performed on the anchor predicted as the ship target.

Similar to detection loss, we also construct the SD loss into three parts: BBox regression with ArBIoU, confidence regression, and classification loss. Thus, the SD loss can be calculated as

$$L_{\mathrm{SD}} = f_{\mathrm{bb}}^{\mathrm{SD}} + f_{\mathrm{cs}}^{\mathrm{SD}} + f_{\mathrm{cla}}^{\mathrm{SD}}. \tag{12}$$

In the distillation process of DSA-Net, the teacher transfers the learned knowledge into the student through enormous duplicate BBoxes, including predictions in the background region labeled with 0, which not only significantly increases the computation of the student model, but also interferes with its learning process. Hence, we add a learning coefficient $C_i^T$ to formulate the SD loss as a confidence-scaled function. This scheme is to encourage the student model only learn useful knowledge when the prediction outputs in the teacher model are not equal to 0. This ensures that the knowledge transfer from teacher to student is not performed in the region without

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information. The learning coefficient can be calculated as

$$C_i^T = \begin{cases} 0, & \widehat{p_{\text{li}}} = 0 \\ 1, & \text{otherwise} \end{cases} \tag{13}$$

where $\widehat{p_{\text{li}}}$ is the semi-soft label for $i$th BBox. Note that the confidence regression does not desire scaling operation since the negative candidates have little impact on it [27]. Therefore, the total BBoxes losses are given by

$$f_{\text{bb}}^{\text{total}}(\boldsymbol{b}^{\boldsymbol{GT}}, \widehat{\boldsymbol{b}}, \widehat{\boldsymbol{b}}^{\boldsymbol{T}}, \widehat{\boldsymbol{b}}^{\boldsymbol{S}}) = \underbrace{f_{\text{bb}}(\boldsymbol{b}^{\boldsymbol{GT}}, \widehat{\boldsymbol{b}})}_{\text{Detection loss}} + \underbrace{f_{\text{bb}}^{\text{SD}}(\widehat{\boldsymbol{b}}^{\boldsymbol{T}}, \widehat{\boldsymbol{b}}^{\boldsymbol{S}})}_{\text{SD loss}} \tag{14}$$

where $\widehat{\boldsymbol{b}}^{\boldsymbol{T}}$ and $\widehat{\boldsymbol{b}}^{\boldsymbol{S}}$ denote the parameter offset vector predicted by the teacher and student networks, respectively. $f_{\text{bb}}^{\text{SD}}$ is the BBox regression in SD, and it can be calculated as

$$f_{\text{bb}}^{\text{SD}}(\widehat{\boldsymbol{b}}^{\boldsymbol{T}}, \widehat{\boldsymbol{b}}^{\boldsymbol{S}}) = \frac{1}{N}\sum_{i=1}^{N} \text{obj}_i \cdot C_i^T \cdot |\log(\text{ArBIoU})| \tag{15}$$

where $C_i^T$ is the binary coefficient for filtering out the redundant BBoxes without "dark knowledge." Similarly, the total classification loss is given by

$$\begin{aligned} &f_{\text{cla}}^{\text{total}}\left(p^{\text{GT}}, \widehat{p}, \widehat{p_l}, \widehat{p}^S\right) \\ &= f_{\text{cla}}\left(p^{\text{GT}}, \widehat{p}\right) + f_{\text{cla}}^{SD}\left(\widehat{p_l}, \widehat{p}^S\right) \end{aligned} \tag{16}$$

$$\begin{aligned} f_{\text{cla}}^{\text{SD}}\left(\widehat{p_l}, \widehat{p}^S\right) = -\frac{1}{N}\sum_{i=1}^{\frac{N}{3}} &\text{obj}_i \cdot C_i^T \\ &\cdot \left[\widehat{p_{\text{li}}} \log\left(\widehat{p_i}^S\right) + \left(1 - \widehat{p_{\text{li}}}\right) \log\left(1 - \widehat{p_i}^S\right)\right] \end{aligned} \tag{17}$$

where $p_l$ is the semi-soft label, and $\widehat{p}^S$ represents the class probability output by student network. $f_{\text{cla}}^{SD}$ represents the classification loss in SD. The total confidence loss is given by

$$f_{\text{cs}}^{\text{total}}(C^{\text{GT}}, \widehat{C}, \widehat{C}_T, \widehat{C}^S) = f_{\text{cs}}(C^{\text{GT}}, \widehat{C}) + f_{\text{cs}}^{\text{SD}}(\widehat{C}_T, \widehat{C}^S) \tag{18}$$

where $\widehat{C}$ and $\widehat{C}^S$ represent the confidence scores calculated by the teacher and student networks, respectively. $f_{\text{cs}}^{\text{SD}}$ is the confidence regression in SD, whose calculation is similar to (9). In conclusion, the final loss function for training can also be formulated as

$$\begin{aligned} L_{\text{Final}} = &f_{\text{bb}}^{\text{total}}(\boldsymbol{b}^{\text{GT}}, \widehat{\boldsymbol{b}}, \widehat{\boldsymbol{b}}^{\boldsymbol{T}}, \widehat{\boldsymbol{b}}^{\boldsymbol{S}}) + f_{\text{cs}}^{\text{total}}(C^{\text{GT}}, \widehat{C}, \widehat{C}_T, \widehat{C}^S) \\ &+ f_{\text{cla}}^{\text{total}}(p^{\text{GT}}, \widehat{p}, \widehat{p_l}, \widehat{p}^S). \end{aligned} \tag{19}$$

## IV. Experiments and Discussion

In this section, we conduct extensive experiments on the open SAR dataset and illustrate the effectiveness of the proposed method. First, the dataset and evaluation metrics in our experiments are introduced, and then the hyperparameter settings are described and analyzed. Next, we focus on the investigation and discussion of the proposed approach, including the influence of ArBIoU loss and the semi-soft label with SD strategy. Finally, we compare the proposed method with existing CNN-based detectors in different scenarios to verify the superiority of our network on a public dataset and prove the generalization ability of the proposed algorithm in a large-scale SAR image.

In our work, the pretrained CSPDarkNet53 on the MS COCO dataset [51] is adopted as the initial backbone network and trained with the learning rate of 0.001. Meanwhile, we utilize the Adam algorithm [52] with four images per mini-batch to optimize the training process. Additionally, the batch size is set as 4, and the epoch number is set as 200. All experiments are implemented with the PyTorch framework, CUDA 11.0, and performed on an NVIDIA Tesla A100 GPU card with 80-GB memory.

### A. Experiential Datasets and Evaluation Metrics

To verify the performance of our proposed approach, the public rotation SAR ship detection dataset (SSDD+) and the large-scale image are collected as the touchstones.

SSDD+ is an extended version of SSDD [53] with extra angle annotation, which has the same target information as SSDD, and facilitates the investigation of rotation-based ship detection in SAR images. This dataset has a total of 1160 images with 2501 ships, which are divided into a training set and a test set with a proportion of 7:2, wherein 34 inshore images and 198 offshore images are included in the test set. Moreover, a large-scale SAR image covering Changyi port, Singapore, is utilized to verify the generalization ability of the proposed method compared with the baseline. Specifically, our method is trained on the training set of SSDD+, then directly tested on the large-scale SAR image of the Changyi port for comparison in visualization.

In addition, five measurements are employed to evaluate the detection effectiveness quantitatively, including the precision, recall, average precision (AP), F1 score, and frames per second (FPS). To calculate these criteria, we predefine the true positive (TP), false positive (FP), false positive (FP), and false negative (FN). Typically, the predicted anchor can be judged as TP when the IOU value is larger than the preset threshold; otherwise, it is an FP. Meanwhile, FP and FN are the numbers of false alarms and undetected ships, respectively. We calculate the recall, precision, and F1 score as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

and AP is the area under the precision-recall curve (PRC)

$$\text{AP} = \int_0^1 P(R) dR \tag{23}$$

where $P$ and $R$ indicate the recall rate and precision rate, respectively.

### B. Hyperparameter Setting

To evaluate the effect of the hyperparameters, a series of experiments are conducted on the training set and then validated on the test set. Specifically, we first perform comparative experiments on the training set under different temperature conditions and test separately on the test set to select the

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TABLE I
INFLUENCE OF DIFFERENT TEMPERATURE VALUES

| $T$ | 1 | 2 | 5 | 10 | **20** | 30 | 50 | 100 | 500 |
|---|---|---|---|---|---|---|---|---|---|
| **AP** | 87.95 | 88.06 | 88.94 | 91.06 | **91.81** | 91.70 | 91.04 | 87.51 | 79.77 |

TABLE II
INFLUENCE OF DIFFERENT CONFIDENCE VALUES

| $T_{Con}$ | 0.1 | 0.2 | **0.3** | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|---|---|---|---|
| **AP** | 91.71 | 91.64 | **91.92** | 91.93 | 91.81 | 91.85 | 91.75 | 91.53 | 88.91 |

TABLE III
INFLUENCE OF DIFFERENT NMS VALUES

| $T_{NMS}$ | 0.1 | **0.2** | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|---|---|---|---|
| **AP** | 88.39 | **93.44** | 92.53 | 91.95 | 91.81 | 89.84 | 85.91 | 69.64 | 56.39 |

TABLE IV
INFLUENCE OF DIFFERENT IOU VALUES

| $T_{IoU}$ | 0.1 | **0.2** | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
|---|---|---|---|---|---|---|---|---|---|
| **AP** | 95.70 | **95.64** | 95.41 | 94.40 | 91.81 | 81.19 | 70.73 | 28.87 | 2.07 |

suitable temperature value. Then, the inference experiments with different values of the confidence, nonmaxima suppression (NMS), and IoU parameters are only performed on the test set at the selected temperature ($T = 20$) to determine the optimal combination of parameters.

$T$ is the SD temperature in the training process, which directly affects the distillation results and is used to smooth labels, i.e., to exploit the negative sample information. Too small a value will lead to unobvious distillation, especially when the high temperature is equal to 1, the detection accuracy is similar to the situation without SD. On the other hand, too large a value of $T$ will cause the network to overemphasize the negative label information, resulting in performance degradation. To motivate the reinforcement effect of SD, the temperature value is set to 20, as given in Table I.

$T_{\text{Con}}$ is the confidence threshold to determine whether the detected area is a true ship in the inference stage. Intuitively, the influence of confidence on AP is not prominent, as shown in Table II. This may be because the confidence scores of the correct ship targets are inherently high. The value is selected as 0.3 in our model.

$T_{\text{NMS}}$ is the NMS threshold to remove duplicate BBoxes from prediction results. In prediction, a low $T_{\text{NMS}}$ threshold is beneficial for eliminating redundant duplicates. However, the too-small $T_{\text{NMS}}$ leads to missed detection in dense inshore scenarios. On the contrary, the too large $T_{\text{NMS}}$ will lead to invalid suppression of redundant candidates. The empirical value is recommended to be 0.2∼0.5, as shown in Table III. In this article, we set the $T_{\text{NMS}}$ threshold as 0.2.

$T_{\text{IoU}}$ is the IoU threshold used to distinguish between positive and negative samples for inference. As presented in Table IV, the smaller the value, the more ship targets can be judged as positive samples. Therefore, the selected value is recommended to be 0.1∼0.3. In our experiments, it is set to 0.2.

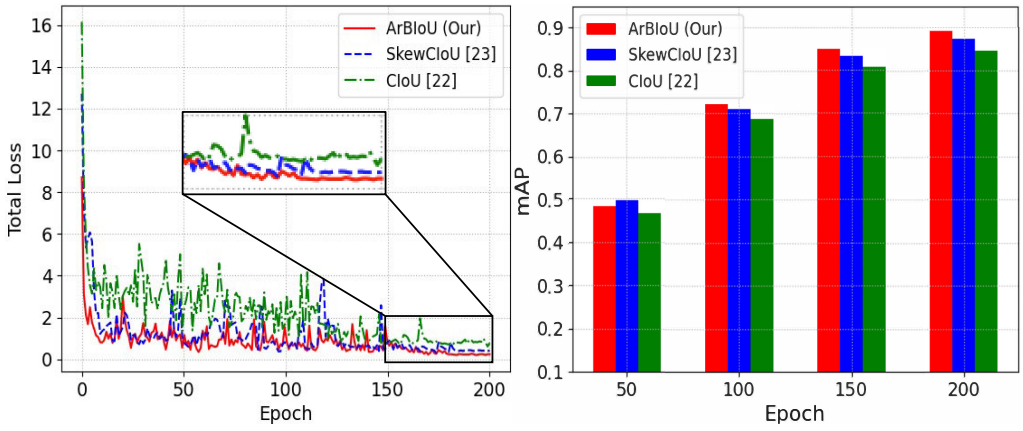
TABLE V
DETECTION RESULTS WITH DIFFERENT COMPONENTS ON SSDD+

| Baseline | ArBIoU loss | Semi-Soft label with SD | $AP_{0.5}$ (%) |
|---|---|---|---|
| DSA-Net [23] | × | × | 88.91 |
| | √ | × | 89.93 (+1.02) |
| | × | √ | 92.25 (+3.34) |
| | √ | √ | **93.56** (+4.65) |

### C. Ablation Study

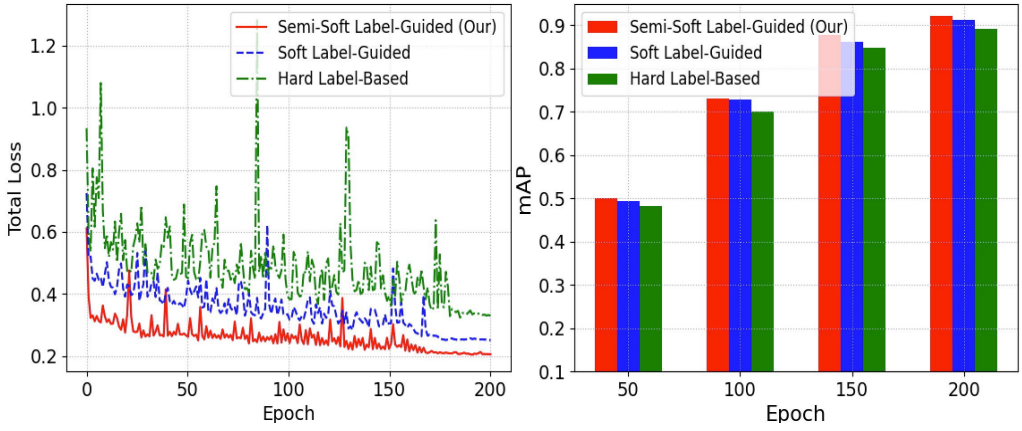
In ablation experiments, we select DSA-Net as the baseline to verify the effectiveness of the proposed components, where the baseline network without ArBIoU loss term is the corresponding network with its original loss version, i.e., SkewCIoU loss term [24]. Two modifications are devised to promote the baseline more properly for the representation of ship targets and the application in complex background. Overall quantitative comparisons are reported in Table V with the same parameter settings to be fair. As we can see, both proposed techniques can improve detection accuracy in inference. Compared with the baseline, the proposed semi-soft label with SD and ArBIoU loss achieve 3.34% and 1.02% higher on $AP_{0.5}$, respectively. Furthermore, $AP_{0.5}$ gains a major enhancement of 4.65% by integrating both SD and ArBIoU loss. In the following parts, the influence of each component in the training process will be elaborated in detail, separately.

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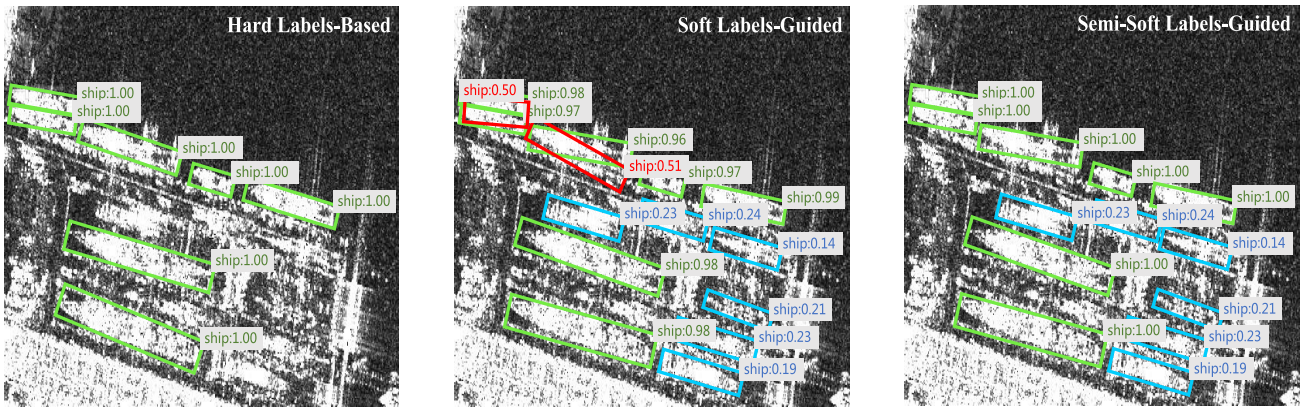

(a) (b)

Fig. 4. Comparison of different losses in DSA-Net. (a) Total loss value during training. (b) Accuracy in different epochs.




(a) (b)

Fig. 5. Effective verification of the proposed semi-soft label. (a) Total loss value during training. (b) Accuracy in different epochs.




(a) (b) (c)

Fig. 6. Visualization of sample distribution in different labels. The green, blue, and red bounding boxes indicate the positive samples, the negative samples, and the mismatched annotations, respectively. (a) Distribution of hard labels. (b) Distribution of soft labels. (c) Distribution of proposed semi-soft labels.

*1) Effect of ArBIoU:* We first investigate the impact of the different losses during the training period. As presented in Fig. 4(a), the proposed ArBIoU outperforms SkewCIoU and CIoU with the most stable training curve and the minimum convergence value. In addition, constrained by the poor quality of SAR images and a limited quantity of positive samples in existing public SAR datasets, deep networks are difficult to adequately and quickly learn the feature information of ship targets by simply modifying the loss function without improving the dataset. Therefore, all the training loss curves are initially not smooth until about the 180th epoch, after which the loss values remain basically unchanged, indicating that the

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TABLE VI
DETECTION RESULTS COMPARISON IN DIFFERENT SCENARIOS ON SSDD+

| Method | Recall (%) | Precision (%) | F1 Score (%) | mAP (%) |
|---|---|---|---|---|
| Offshore: | | | | |
| Rotated-RetinaNet [46] | 85.60 | 69.63 | 76.79 | 90.80 |
| CenterNet-OBB [33] | 81.61 | 32.19 | 46.17 | 90.88 |
| Faster RCNN-Oriented [30] | 90.53 | 44.50 | 59.67 | 91.15 |
| ROI transformer [47] | 50.07 | **95.07** | 65.59 | 92.27 |
| FCOS [34] | 90.85 | 66.39 | 76.72 | 92.55 |
| $R^2$CNN [35] | 89.69 | 43.98 | 59.01 | 94.63 |
| YOLOv4-Oriented [48] | 95.58 | 62.45 | 75.59 | 94.78 |
| $R^3$Det [40] | 91.33 | 63.68 | 75.04 | 97.56 |
| **$S^2$LSDNet (Our)** | **99.43** | 75.00 | **85.50** | **99.20** |
| Inshore: | | | | |
| CenterNet-OBB [33] | 80.42 | 29.51 | 43.18 | 79.81 |
| Rotated-RetinaNet [46] | 87.32 | 54.17 | 66.86 | 81.03 |
| YOLOv4-Oriented [48] | 85.03 | 52.66 | 65.04 | 82.33 |
| ROI transformer [47] | 78.33 | 60.14 | 68.04 | 83.25 |
| Faster RCNN-Oriented [30] | 79.41 | 50.37 | 61.64 | 83.88 |
| FCOS [34] | 88.37 | 45.11 | 69.73 | 84.11 |
| $R^3$Det [40] | 80.32 | 58.18 | 67.48 | 84.37 |
| $R^2$CNN [35] | 82.19 | 37.77 | 51.76 | 85.63 |
| **$S^2$LSDNet (Our)** | **93.58** | **65.38** | **76.98** | **89.26** |

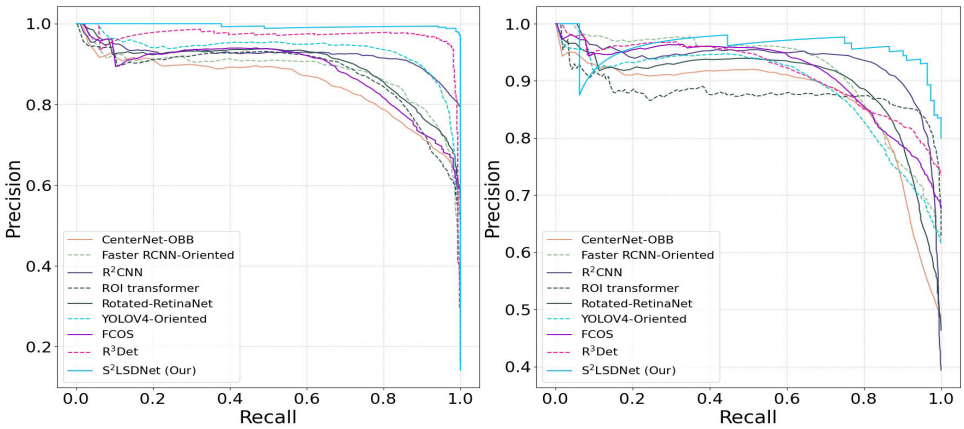



Fig. 7. PR curves of different methods in inshore and offshore scenarios. (a) PR curves in offshore scenario. (b) PR curves in inshore scenario.

models have reached convergence. Intuitively, the loss curve of the proposed ArBIoU has minimal oscillations compared with other loss criteria during the training process. Therefore, the proposed method has better data fitting ability. Moreover, the detection results in different epochs with the above-mentioned three criteria are shown in Fig. 4(b). In comparison with the other two-loss criteria, the proposed ArBIoU-based DSA-Net basically maintained the highest detection accuracy throughout the entire process, except that the results were slightly inferior to that of SkewCIoU in the 50th epoch. This is because the networks are not in convergence at the initial phase. Consequently, the ArBIoU loss is beneficial for network training in SAR ship detection due to the design of the weights of aspect ratio offset and center point bias in the BBoxes regression.

*2) Effect of the Semi-Soft Label With SD:* The training loss curves of different label-guided methods are displayed in Fig. 5(a). Compared with the models guided by hard labels and soft labels, the proposed semi-soft label-guided model yields excellent and stable performance in the training process.

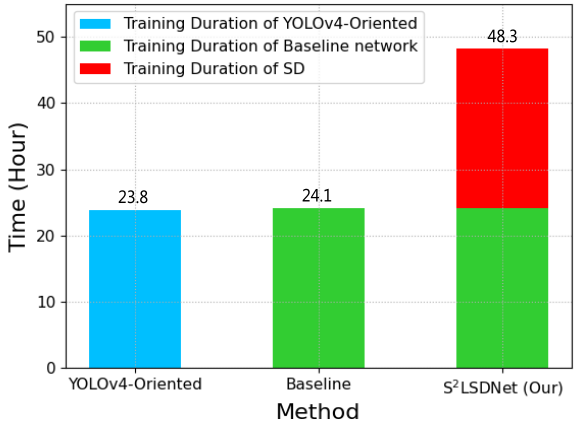
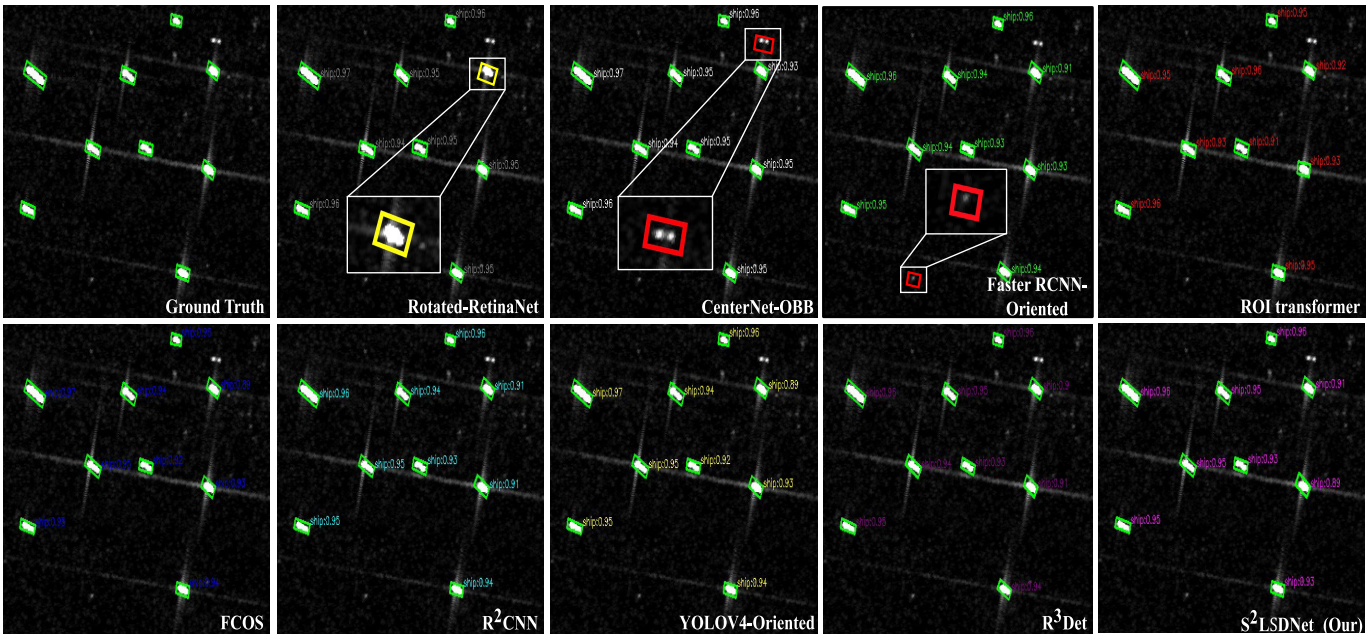


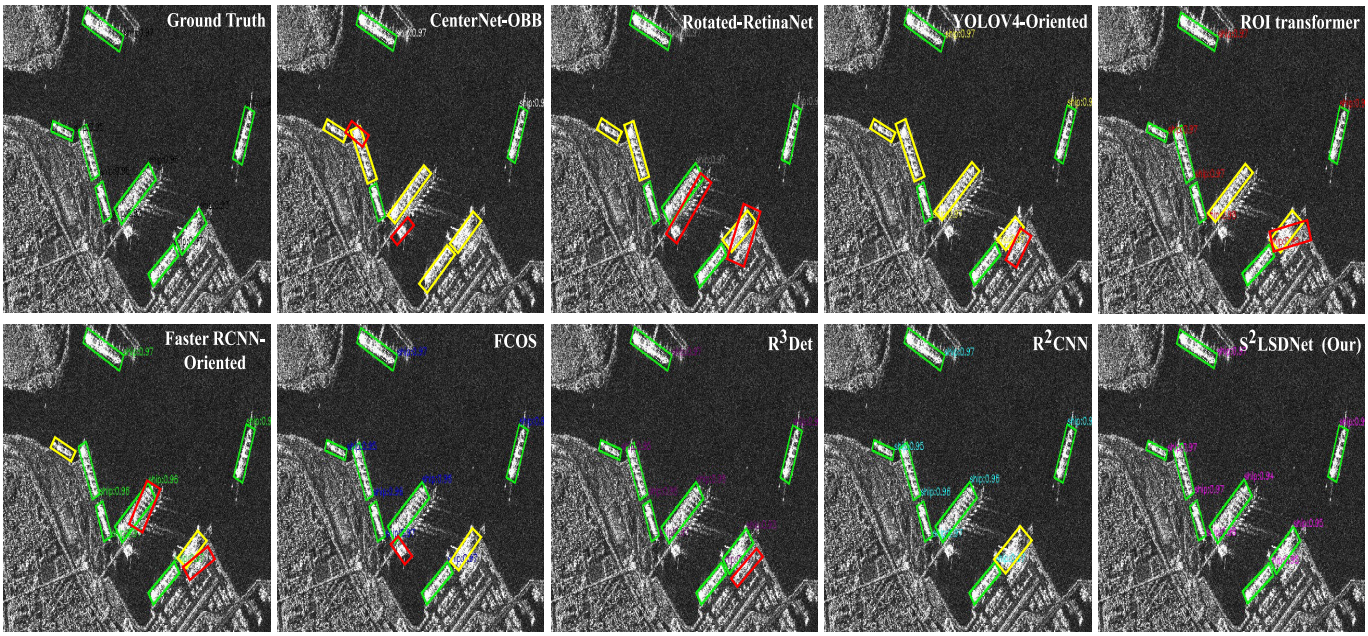

Fig. 8. Comparison of training times for different methods. The baseline is the improved DSA-Net [24] with ArBIoU loss. The height of the column indicates the total training time in hours.

Meanwhile, our method improves the detection accuracy by a large margin during the testing phase, as shown in Fig. 5(b). Additionally, the visualization with different labels is given

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(a)




(b)

Fig. 9. Visualization instance of SOTA methods in different scenes on SSDD+. The green, red, and yellow rectangles indicate the correctly detected ship, false alarms, and missed detection, respectively. (a) Ship detection results in the offshore scene. (b) Ship detection results with inshore cases.

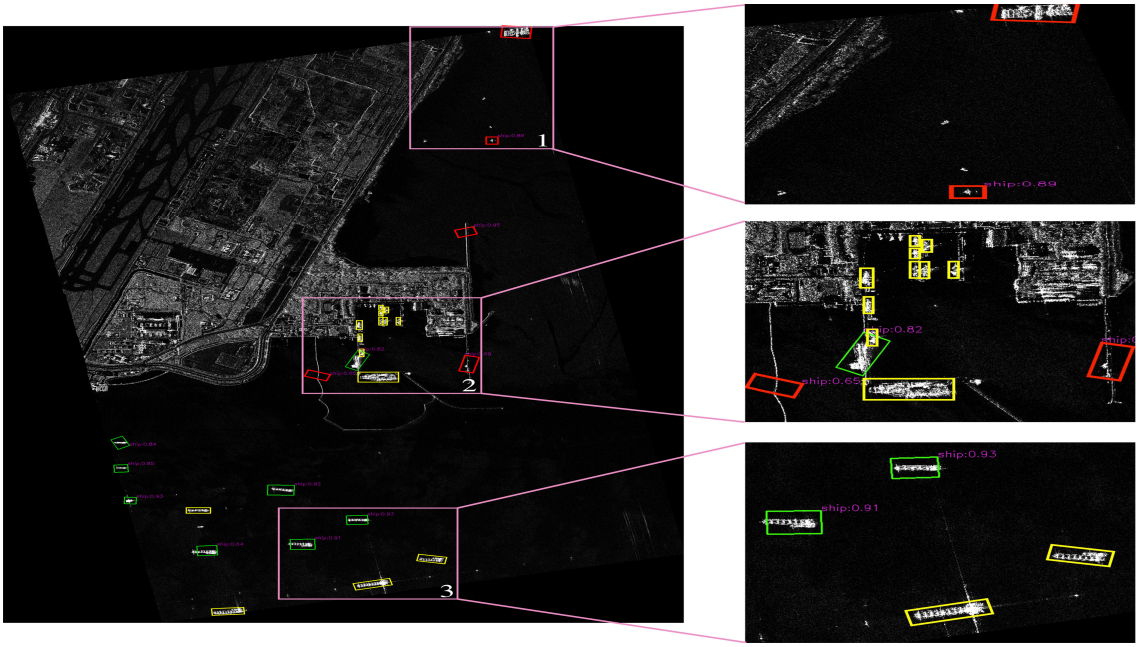
in Fig. 6. It can be seen that the proposed semi-soft label-guided model can leverage the instructive information in positive samples and "dark knowledge" from negative samples without mismatched annotations to improve detection accuracy, compared with two other methods.

### *D. Detection Performance Comparisons*

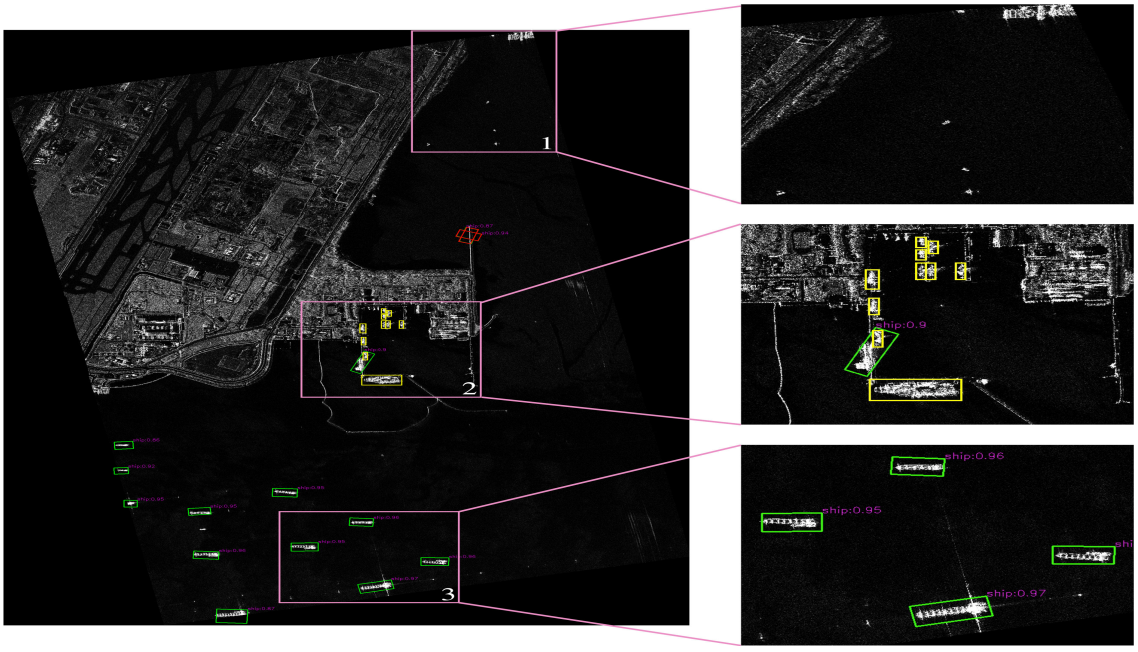
We further analyze the proposed S$^2$LSDNet on the SSDD+ dataset in this subsection. First, we conduct accuracy comparisons to show the efficacy of our method in different scenes. Then, we present the detection performance of our method using qualitative experiments on total images in the test set. Note that we select the optimal combinations of hyperparameters for comparison methods as given by their original publications.

Different from offshore ship detection, ship detection in inshore environments is more challenging due to the substantial interference caused by complex background clutter. Hence, the experiments are employed, respectively, in the offshore and inshore scenes to evaluate the performance of our network. Intuitively, the proposed S$^2$LSDNet can achieve a significant advantage in both scenarios, as shown in Table VI.

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(a)




(b)

Fig. 10. Visualization of detection results in a large-scale SAR image. The green, red, and yellow rectangles indicate the correctly detected ship by network, false alarms, and missed detection, respectively. (a) Detection results of the DSA-Net with ArBIoU. (b) Detection results of the proposed $S^2$LSDNet.

Specifically, our method acquires improvements on the indicators of recall, F1 score, and mAP in the offshore scene, especially with an increase of 1.64% on mAP compared with the SOTA detector. In terms of inshore ship detection, the proposed model improves the overall performance by a considerable margin, especially for a gain of 3.63% on mAP. Additionally, the corresponding PR curves in different scenes are displayed in Fig. 7. It can be seen that the proposed $S^2$LSDNet achieves better results with the maximum enclosed area compared with other advanced rotation detectors. Fig. 8 illustrates the comparison of training times for different networks, where the training time of YOLOv4-Oriented with the fastest test speed among the commonly used methods is 23.8 hours, which is comparable with the baseline network with 24.1 hours. Intuitively, the training duration of our method is twice that of the baseline due to the additional SD operation. In Fig. 9, several detection visualizations in different scenes are presented to illustrate the performance of different methods. It can be observed from Fig. 9(a) that most of the existing methods have obtained tolerable

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TABLE VII
DETECTION PERFORMANCE ON SSDD+

| Framework | Method | Backbone | Image Size | mAP (%) | Test Time (FPS) |
|---|---|---|---|---|---|
| Anchor-Free | CenterNet-OBB [33] | DLA-34 | 512×512 | 87.45 | 21.8 |
| | FCOS [34] | MobileNetV2 | 1024×1024 | 90.83 | 31.9 |
| | FBR-Net [36] | ResNet50 | 512×512 | 94.10 | 24.0 |
| | STP-Net [11] | ResNet101 | 800×800 | 94.89 | — |
| | SKG-Net [10] | ResNet50 | 512×512 | 95.02 | 16.0 |
| | MGF-CenterNet [49] | ResNet50 | 512×512 | 96.17 | <21.8 |
| | KC-CenterNet [38] | DLA-34 | 512×512 | 96.30 | 26.0 |
| Anchor-Based | Rotated-RetinaNet [46] | ResNet101 | 800×800 | 87.41 | 21.1 |
| | YOLOv4-Oriented [48] | CSPDarkNet53 | 608×608 | 90.20 | 35.1 |
| | ROI transformer [47] | ResNet101 | 512×800 | 90.39 | 11.2 |
| | Faster RCNN-Oriented [30] | ResNet50 | 512×512 | 90.80 | 18.6 |
| | $R^2$CNN [35] | ResNet101 | 800×800 | 92.89 | 10.3 |
| | $R^3$Det [40] | ResNet101 | 800×800 | 93.33 | 28.9 |
| | MD Faster RCNN [50] | ResNet50 | 512×512 | 96.20 | <18.6 |
| | **$S^2$LSDNet (Our)** | CSPDarkNet53 | 608×608 | **97.08** | 30.5 |
| | **Tiny-$S^2$LSDNet (Our)** | Tiny CSPDarkNet53 | 608×608 | 93.42 | **61.6** |

detection performance in the offshore scenario, indicating that the bottleneck restricting the ship detection accuracy mainly comes from the complex inshore scene. Therefore, our work focuses on inshore ship detection. In the inshore scenario, our network outperforms other competitors with better accuracy benefiting from the full utilization of the semi-soft label generated by the proposed SD technique and the ArBIoU criterion, as shown in Fig. 9(b). Additionally, the overall detection results on the SSDD+ dataset are illustrated in Table VII. It can be seen that the proposed $S^2$LSDNet has achieved higher accuracy, i.e., 97.08%, on mAP compared with other SOTAs. Regarding the test time comparison as shown in Table VII, our method obtains a satisfactory computation-efficient with anterior ranking for speed (about 30.5 FPS). Furthermore, we halve the channel number of the baseline model implemented by the channelwise pruning technique and train it with the proposed SD strategy to obtain a slimming student network (i.e., Tiny-$S^2$LSDNet). As shown in Table VII, the proposed Tiny-$S^2$LSDNet can achieve a faster speed of 61.6 FPS without remarkable performance degradation. In conclusion, our proposed approach achieves a superior performance to existing commonly used detectors on public SSDD+ datasets with tolerable computational complexity.

### *E. Generalization Performance Verification in Large-Scale SAR Image*

In order to verify the robustness and generalization ability of the proposed semi-soft label with SD technique, $S^2$LSDNet and the baseline detector are tested on a large-scale SAR image of the Changyi port, Singapore, without participation in the training, as displayed in Fig. 10. It can be observed that the proposed method has superior generalization performance and stronger antiinterference capability with more correct detections and fewer errors compared with the baseline. Specifically, we focus on the detection results in the image from three areas for demonstration. The first area presents the inshore scenario. Unlike appearing several false alarms in the baseline, the proposed $S^2$LSDNet shows fewer errors in suppressing environmental interference. The second area is captured for the complex inshore scene with multiscale ship distribution. Despite the fewer false alarms in our scheme, both models suffer from a few missed detections, especially for the small ship targets near the shore, indicating that the proposed semi-soft label with SD has limited benefit for multiscale ship detection in the inshore scenario. The third area presents an offshore scenario. Intuitively, our method makes the correct detections, while the baseline has missed detections under the influence of poor generalization ability.

To sum up, the proposed approach achieves better detection accuracy than existing methods, especially in inshore cases with the assistance of semi-soft labels generated by the SD strategy. However, several limitations still exist in our work. The issues of multiscale ship detection in complex scenes and accelerated detection methods will be investigated in our future work.

## V. CONCLUSION

This article proposes a novel semi-soft label-guided network based on SD for SAR ship detection against inshore background clutter. With the aim to exploit the negative sample information, an SD strategy in different temperatures is devised to boost detection accuracy through the utilization of semi-soft labels. In addition, we develop an ArBIoU loss for accurate regression based on quantifying the interaction weights of aspect ratio difference and center point offset in ship detection. The ablation studies manifest that the proposed components play a crucial role in performance improvement. Substantial experimental comparisons with commonly used CNN-based detectors on the public SSDD+ dataset illustrate the superiority of our method, and the generalization performance of our method is demonstrated by using a large-scale SAR image out of the distribution of the SSDD+ dataset.

Furthermore, the multiscale problem is also a key task in SAR ship detection, while the anchor-free framework has been proven to be suitable for such scenes. Therefore, we will explore an improved anchor-free network combined with SD to tackle the problem of multiscale in SAR ship detection in our future research.

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